Predicting Airbnb Listing Price Across Different Cities

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UE24CS352A Machine Learning, Mini Project 71 · Reference: Luo, Zhou & Zhou, Stanford CS229 (2019)

1. Problem statement

Airbnb hosts set their nightly price by guesswork, and guests cannot easily tell whether a price is fair. We predict the nightly price of a listing from its attributes (size, room type, location, amenities, reviews, host details and description) in three cities, **New York City, Paris and Berlin**, and ask two further questions: does one model trained on several cities work as well as separate models, and can a model trained in some cities price listings in a city it has never seen?

2. Dataset

We use the public **Inside Airbnb** detailed listings files (scraped June/July 2026; 90 columns per listing). Listings without a price were removed, as were prices outside 10 to 1000 USD per night (1.3% to 5% per city). Paris and Berlin prices (EUR) were converted to USD at a fixed 1.15. Each city was split 80/20 into train and test with a fixed seed, and every experiment is scored on the same test listings.

City	Raw listings	With price	Used	Train / test	Median price	Min. stay ≥ 28 nights
New York City	30,259	21,515	20,890	16,712 / 4,178	$170	78%
Paris	77,679	48,402	46,000	36,800 / 9,200	$227	14%
Berlin	12,776	8,441	8,332	6,665 / 1,667	$151	20%

Table 1. Data per city (prices in USD after conversion).

3. Approach

- **Target.** Prices are strongly right-skewed, so all models predict **log(price)**; as in the reference report this was the single most useful transformation.
- **Features (189 in total), in groups.** *C, continuous*: accommodates, bedrooms, beds, bathrooms (parsed from text) and a shared-bath flag, minimum/maximum nights, availability, review counts and the seven review scores, host tenure and listing counts, superhost and licence flags, amenity count, description length, *distance to the city centre* and latitude/longitude; median-imputed with missing-value flags and standardised. *O, categorical*: room type and property type (one-hot) and the 60 most common amenities (multi-hot). *T, text*: TF-IDF of title and description on uni- and bi-grams, reduced to 50 dimensions by truncated SVD.
- **Leakage control.** Columns derived from the price itself (`price_quote_*`, `estimated_revenue_l365d`) and the length of the stay that was quoted were excluded.
- **Models.** Ridge regression and K-nearest neighbours (baselines), Random Forest, XGBoost (hyper-parameters chosen by 3-fold random search on the NYC training set, CV R^2 0.848) and a neural network (MLP, 64 and 32 ReLU units, Adam, early stopping). **Metrics:** R^2 and MSE on log price, plus MAE and median percentage error in dollars.

4. Implementation

Python with pandas, scikit-learn and XGBoost. The whole feature pipeline is a single scikit-learn `ColumnTransformer` (custom transformers for amenities and text), so the same object is fitted on training data only and reused for prediction, which prevents test-set leakage. `scripts/prepare_data.py` cleans the raw files, `scripts/train.py` runs all four experiments and writes metrics and figures, and the final model (XGBoost on all three cities with a city indicator) is saved and served by a command-line predictor and a Streamlit web app for the demo. A Jupyter notebook walks through the analysis.

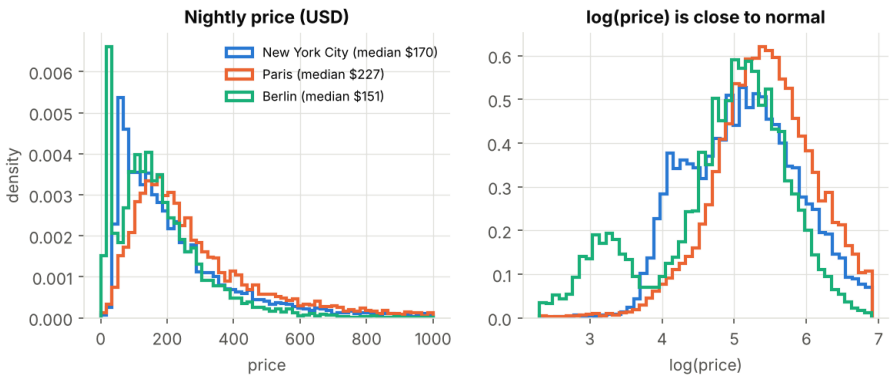


Figure 1. Price distribution per city. Raw prices are skewed; log prices are close to normal. Berlin's low-price bump is long-stay listings quoted at monthly rates.

5. Results

E1: model (all features)	NYC	Paris	Berlin	MAE NYC
Ridge regression	0.769	0.707	0.865	$64
K-nearest neighbours	0.760	0.651	0.807	$64
Random Forest	0.825	0.799	0.874	$55
XGBoost	**0.850**	**0.838**	**0.895**	**$50**
Neural network (MLP)	0.797	0.798	0.851	$60

Table 2. Test R^2 on log price per city (MAE in USD for NYC).

E2: XGBoost features	C	O	T	C+O	C+O+T
NYC	0.843	0.556	0.532	0.856	0.850
Paris	0.817	0.379	0.375	0.833	0.838

Table 3. Feature-group ablation (test R^2, log price).

E3/E4: trained on	NYC	Paris	Berlin
NYC only	0.844	0.114	0.226
Paris only	0.484	0.829	**0.770**
NYC + Paris	0.836	0.825	0.694
Berlin only	−0.808	0.495	0.892

Table 4. Cross-city test R^2 (XGBoost, no lat/lon or city features). Berlin is never seen by the first three models.

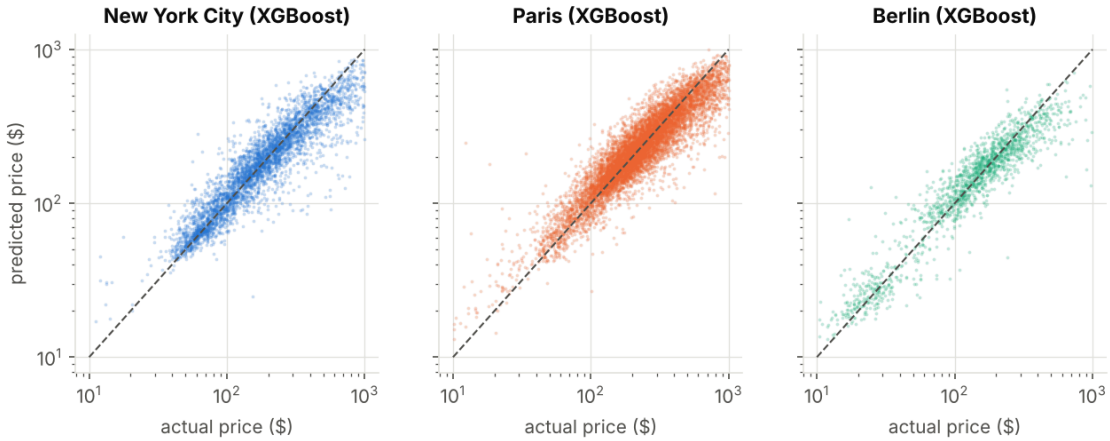


Figure 2. XGBoost predicted vs actual nightly price on each city's test set (log scales).

- **XGBoost is the best model in every city** (test R^2 0.850 NYC, 0.838 Paris, 0.895 Berlin; median error about 15% of the price), ahead of Random Forest and the MLP; the linear and KNN baselines trail by 0.03 to 0.19. The reference report reached 0.741 (NYC) and 0.704 (Paris).
- **Structured features carry the signal.** Continuous features alone reach 0.84/0.82; categorical and amenity features add about 0.015, and the description text adds almost nothing on top because it mostly restates the structured fields. The strongest single features are the shared-bath flag, licence status, room type, guest capacity and minimum stay.
- **One model for two cities** (NYC + Paris) is nearly as accurate as two separate models (0.836 vs 0.844 and 0.825 vs 0.829), so a single multi-city model is practical.
- **Transfer depends on market similarity, not data size.** A Paris model prices unseen Berlin listings with R^2 0.770, but a NYC model only 0.226, and adding NYC to Paris lowers Berlin accuracy to 0.694. Since NYC's 2023 short-term rental law, 78% of its listings require stays of 28+ nights, which are quoted at discounted monthly rates, so NYC prices follow different rules from the European cities. This differs from the reference report, where combining cities helped.

6. Conclusions

Gradient-boosted trees on log price, with carefully cleaned structured features, predict Airbnb prices well in all three cities (R^2 0.84 to 0.90 on log price, typical error about 15%), clearly improving on linear, KNN and neural-network models and on the 2019 reference results. A single model can serve several cities, and a model from a similar market (Paris) generalises to a new city (Berlin) far better than one from a market with different regulation (NYC). Limits: prices are quoted rates at scrape time rather than booked prices, seasonality is not modelled, and a fixed exchange rate is used. Future work: add date and seasonality features, calendar data, and domain adaptation (e.g. per-city re-weighting) to improve transfer to new cities.